



National University of Sciences & Technology (NUST)
School of Electrical Engineering and Computer Science (SEecs)
Department of Computing (DOC)

LEARN | EXPLORE | SOLVE | CREATE

Mathematical Methods for Computing

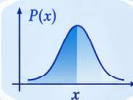
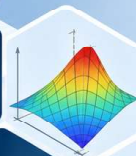
CS-813 | Fall 2026

MSCS Program
SEecs, NUST

Prof. Dr. Quanita Kiran

*Mathematics
Drives
Computing
Forward*

$$Ax = b$$
$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}$$



$$\min_{\theta} f(\theta)$$



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A BRIGHTER TOMORROW

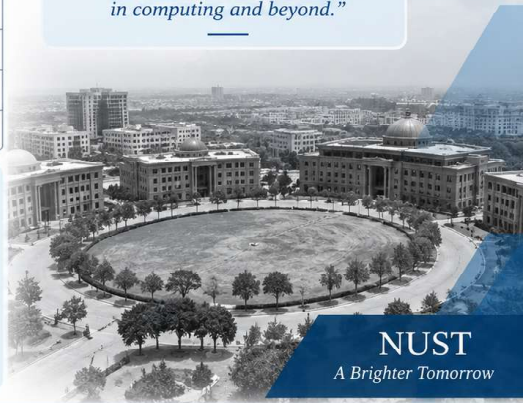
Course Information

Mathematical Methods for Computing (CS-813)

Learn
Explore
Solve
Create

Course Code	CS-813
Semester	Fall 2026
Credit Hours	3+0
Prerequisites	Nil
Class	MSCS-A 2026
Knowledge Group	Core Computer Science

*"A solid mathematical foundation
opens the door to innovation
in computing and beyond."*



Instructor: Prof. Dr. Quanita Kiran



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Lecture Days: Monday, Tuesday



Class Room: CR-12 UG Block SEECS



Consulting Hours: TBD

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SCIENCE | INNOVATION | LEADERSHIP

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Why Mathematical Methods for Computing?

From
Mathematics to
Intelligent Solutions

What is actually happening underneath an AI or Machine Learning algorithm?

1 Real-World Data



Images, text, signals,
networks, measurements...

2 Mathematical Representation

$$X = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & \cdots & x_{mn} \end{bmatrix}$$

Vectors, matrices, graphs,
probability distributions...

3 Computation (Algorithm)



Matrix operations,
optimization, iterative
methods...

4 Model (Learning)



Parameters learned from
data using mathematical
principles

5 Decision (Prediction/Insight)



Classification, prediction,
recommendation,
understanding...

The Mathematical Foundations

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

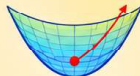
Linear Algebra

Representing and
transforming data



Probability & Statistics

Reasoning under
uncertainty



Optimization

Finding the best
parameters



Matrix Decomposition

Discovering hidden structure
and reducing complexity



Libraries execute algorithms. *Mathematics tells us why they work.*

Meet Your Instructor

Guiding Your Journey in Mathematical Methods for Computing

*Mathematics
for a Smarter
Tomorrow*

Dr. Quanita Kiran

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Adjunct Faculty

SNS-NUST, CEME-NUST

“

*Mathematics is the language
that turns curiosity into discovery.*

”

Research Areas

Exploring Mathematics for Real-World Impact



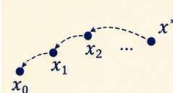
Graph Theory

Modeling and analyzing
complex networks



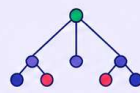
Functional Analysis

Infinite-dimensional spaces
and their applications



Fixed Point Theory

Theoretical foundations
and iterative methods



Discrete Mathematical Structures

Foundations for computing
and algorithm design



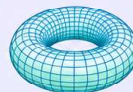
Complex Network Analysis

Structure, dynamics and
applications of real-world
networks



Graph Based Cryptography

Secure communication
using graph structures



Topological Data Analysis

Geometry and topology
in data science



Data Analysis through Graphs

Extracting insights from
data using graph-based
methods

Mathematics connects theory, computation and real-world problems.



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Mathematical Methods for Computing (CS-813)

Course Information

Assessment, Resources and LMS

*Mathematics
for a Smarter
Tomorrow*



Assessment Scheme

Assessment Type	Weightage Range (%)
Assignments	5.0 – 10.0
Project	0.0 – 20.0
Mid Term	25.0 – 35.0
Quiz	10.0 – 15.0
Final Term	40.0 – 50.0
Total	100



Books

Text Books

1. Linear Algebra and Its Applications, 4th Edition, David C. Lay.
2. Mathematics for Machine Learning, Deisenroth et al., 2019.

Reference Books

Additional recommended books and reading material will be shared on LMS during the semester.



Enrollment Code for LMS



Course Enrollment Code:

246195083

Please enroll yourself on the LMS using the above code to access course material, announcements and submissions.



*“Good resources
build strong foundations
for great ideas.”*



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Mathematical Methods for Computing (CS-813)

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Section 1

Linear Algebra

The Language for Representing and Transforming Data

From
Data to
Intelligence

1 What is Linear Algebra?

Linear algebra studies vectors, matrices and related operations to represent, manipulate and analyse data.

“Data in many forms can be represented as vectors and matrices, and linear algebra gives us the tools to work with them.”

2 Why Does It Matter?



Universal representation

From images and text to networks and sensor data – all can be written as vectors or matrices.



Efficient computation

Enables scalable algorithms for large data and AI.



Reveals structure

Helps find patterns, reduce dimensions and make better decisions.



Foundation for ML

Core to learning algorithms such as PCA, regression and neural networks.

3 Data Representation

Real-world data can be represented as vectors or matrices.



Image (grayscale)

$$\begin{bmatrix} 120 & 98 & 102 \\ 85 & 90 & 88 \\ 78 & 92 & 95 \end{bmatrix}$$


Text (bag-of-words)

$$\begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$


Audio/Signal

$$\begin{bmatrix} 0.2 \\ -0.1 \\ 0.4 \\ \vdots \end{bmatrix}$$


Graph/Network

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

4 Topics Covered in This Section

Vectors and Matrices

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \end{bmatrix}$$

- Basic operations (addition, scalar multiplication)
- Matrix operations (multiplication, transpose)
- Types of matrices (identity, diagonal, symmetric)

Systems of Linear Equations

$$\begin{cases} ax + by = c \\ dx + ey = f \end{cases}$$

- Solutions and solution methods
- Gaussian elimination
- Matrix formulation
- Consistency and rank

Determinants and Inverse

$$\det(A), A^{-1}$$

- Determinant and its properties
- Inverse of a matrix
- Methods of computation
- Solving linear systems
- Applications

Inner Product Spaces and Projections

$$\langle x, y \rangle = x^T y$$

- Dot product and norms
- Orthogonality
- Projections
- Least squares
- Applications in data analysis

Eigenvalues and Eigenvectors

$$Ax = \lambda x$$

- Eigenvalues and eigenvectors
- Characteristic equation
- Diagonalization (concept)
- Applications



“Linear algebra turns data into structure, and structure into insight.”



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Section 1

Linear Algebra – Key Applications

From Mathematical Concepts to Real-World Impact

The
Mathematics
Behind
Intelligent
Systems

Why Applications Matter?



Turns data into meaningful information



Core tool in modern computing and AI



Helps us solve real-world problems efficiently

Data Representation with Matrices



Image
($m \times n$ matrix)



Text
(term $\times$ document matrix)



Audio/Signal
(samples $\times$ time)



Graph
(adjacency matrix)

Real-world data in many forms can be represented as matrices.

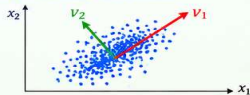
1 Eigenvalue Decomposition (EVD)

Reveals the intrinsic structure of data.

$$A = V\Lambda V^{-1}$$

- Eigenvalues show the amount of variation along each direction.
- Eigenvectors give the principal directions of the data.
- Useful for understanding dynamics, stability and for dimensionality reduction (e.g., PCA).

Example: Principal Directions



Eigenvectors show the main directions of variation in the data.

2 Singular Value Decomposition (SVD)

The most powerful and widely used matrix factorization.

$$A = U\Sigma V^T$$

- Works for any matrix (rectangular or square).
- Decomposes data into orthogonal components.
- Reveals the most important patterns through singular values.
- Widely used in compression, noise removal, recommendation systems and many machine learning algorithms.

$$\begin{matrix} A & U & \Sigma & V^T \\ (m \times n) & (m \times m) & (m \times n) & (n \times n) \end{matrix}$$

Decomposes a matrix into orthogonal components.



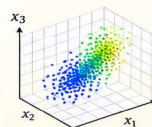
SVD allows us to approximate data using only the most important components, enabling compression and denoising.

3 Principal Component Analysis (PCA)

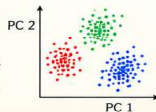
A popular application of EVD/SVD for dimensionality reduction.

- Finds new uncorrelated variables (principal components).
- Captures maximum variance in fewer dimensions.
- Helps visualize, compress and analyze high-dimensional data.
- Used in data visualization, feature extraction and preprocessing for machine learning.

High-Dimensional Data



Low-Dimensional Representation



PCA
Reduce dimensionality

PCA projects data onto the most informative components.

“ Linear algebra is not just theory — it is the foundation that turns data into insight. ”



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Section 1

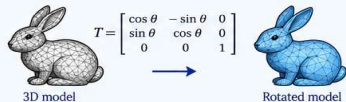
Linear Algebra – Real World Applications

A Common Language Across Science, Engineering and AI

From
Data to
Intelligence

1 Computer Graphics & 3D Vision

Transform, rotate and project objects using matrices.

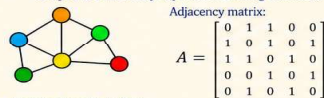


- Geometric transformations (rotation, scaling, translation)
- 3D modelling, animation and rendering
- Camera and perspective projections
- Computer vision (pose estimation, reconstruction)

Matrices turn geometry into computation.

4 Networks and Graphs

Graphs are naturally represented using matrices.

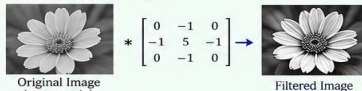


- Social network analysis
- Community detection and link prediction
- Network dynamics (spreading, diffusion)
- Web search and ranking (PageRank)
- Graph-based learning and knowledge graphs

Matrices reveal the structure and patterns in complex networks.

2 Image and Signal Processing

Represent, filter and enhance signals and images using linear algebra.

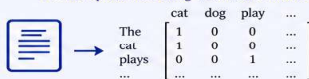


- Image enhancement and filtering (convolution as matrix operation)
- Edge detection and feature extraction
- Audio and signal processing (filtering, noise reduction)
- Medical imaging (MRI, CT, ultrasound)

Signals and images are matrices — linear algebra helps us extract what matters.

5 Natural Language Processing

Text is represented using vectors and matrices.



- Word embeddings (words as vectors)
- Semantic similarity (vector operations)
- Topic modelling and document analysis
- Transformer models (built on matrix computations)
- Information retrieval and search

Words to vectors — linear algebra captures meaning.

3 Machine Learning & AI

Many learning algorithms rely on linear algebra.

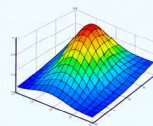


- Linear regression and classification
- Neural networks (matrix multiplications)
- Word embeddings and language models
- Recommendation systems (matrix factorization)
- Deep learning training (optimization with gradients)

From data to predictions — linear algebra is at the core of modern AI.

6 Scientific and Engineering Applications

Linear algebra helps solve real-world problems.



$$Ax = b$$

- Solving large systems of linear equations
- Modelling physical systems (e.g., heat flow, vibrations)
- Control systems and state-space models
- Electrical circuits and power systems
- Robotics and kinematics (transformations)
- Data analysis in scientific simulations

From models to solutions — linear algebra drives scientific discovery.

“Linear algebra is a universal language that turns data into understanding, models into solutions, and ideas into impact.”



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Mathematical Methods for Computing (CS-813)
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Section 2

Vector Calculus and Optimization

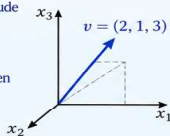
From Change and Direction to Optimal Solutions

Tools
for Modern
Data Science
and AI

1 Vectors and Vector Spaces

Represent and manipulate data using vectors.

- Vector representation, magnitude and direction
- Vector operations (addition, scalar multiplication)
- Dot product and angle between vectors
- Norms and unit vectors
- Vector spaces and bases (brief)



$$\|v\| = \sqrt{v_1^2 + v_2^2 + v_3^2}$$

2 Distances and Similarity Measures

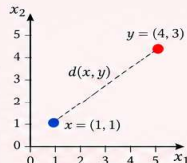
Quantify how similar or different data points are.

- Euclidean distance
- Manhattan distance
- Cosine similarity $\cos(x, y)$
- Other measures (e.g., Jaccard)

$$d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2} \quad (\text{Euclidean})$$

$$d(x, y) = \sum_{i=1}^n |x_i - y_i| \quad (\text{Manhattan})$$

$$\cos(x, y) = \frac{x \cdot y}{\|x\| \|y\|} \quad (\text{Cosine similarity})$$



3 Multivariable Calculus

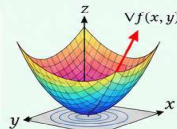
Understand how functions change in multiple dimensions.

- Functions of several variables
- Partial derivatives and gradient
- Directional derivatives
- Gradient and level curves
- Tangent planes (linear approximation)

$$f(x, y) = x^2 + y^2$$

$$\nabla f(x, y) = \begin{bmatrix} 2x \\ 2y \end{bmatrix}$$

$$D_u f(x) = \nabla f(x) \cdot u$$



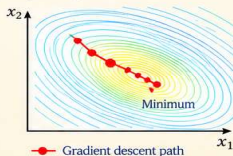
4 Gradient Descent and Optimization

Find the best solution by iteratively improving a function.

- Optimization problems (minimization/maximization)
- Gradient descent algorithm
- Learning rate and convergence
- Convex vs. non-convex functions
- Other methods (e.g., Newton's method, momentum, stochastic gradient descent)

$$x_{k+1} = x_k - \eta \nabla f(x_k)$$

η : learning rate



5 Applications in Computing and AI

Vectors, derivatives and optimization enable many core tasks.



Machine Learning

- Model training (optimization)
- Feature learning (gradients)



Computer Vision

- Image gradients
- Feature extraction



Natural Language Processing

- Word embeddings
- Similarity search



Robotics and Control

- Trajectory optimization
- Control and state estimation



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Section 3

Probability and Statistics

Reasoning Under Uncertainty

From
Data to
Decisions

1 Why Probability and Statistics?

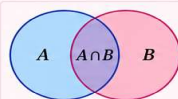
Real-world data is uncertain, noisy and incomplete.

- Probability helps us model randomness and reason under uncertainty.
- Statistics helps us learn from data, make inferences and draw conclusions.
- **Essential** for machine learning, pattern recognition, experiment design and decision making.

Data is everywhere — probability and statistics help us make sense of it.

2 Key Concepts in Probability

- Sample space and events
- Axioms of probability
- Conditional probability
- Bayes' theorem
- Random variables (discrete and continuous)
- Expectation and variance
- Common distributions (Binomial, Poisson, Normal)

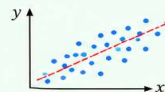


$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Bayes' theorem helps us update our beliefs with new evidence.

3 Key Concepts in Statistics

- Descriptive statistics (mean, median, variance)
- Estimation (point and interval)
- Hypothesis testing
- Regression analysis
- Correlation and covariance
- Maximum likelihood estimation
- Applications in data science and ML

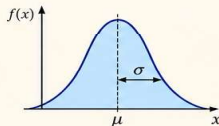


Linear regression model:
 $y = \beta_0 + \beta_1 x + \varepsilon$
Used to model and predict relationships in data.

4 Example: Normal Distribution

Many real-world measurements (e.g., height, error, noise) follow a normal distribution:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$



- μ : mean (center of the distribution)
- σ : standard deviation (spread)
- Symmetric and bell-shaped curve.
- Widely used in statistics, ML and science.

The normal distribution models natural variation in many real-world phenomena.

5 Real-World Applications

Machine Learning
Model uncertainty, classification, prediction

Finance
Portfolio modeling, market analysis

Healthcare
Risk assessment, clinical trials

Social Sciences
Surveys, A/B testing, opinion analysis

“ Probability quantifies uncertainty. Statistics turns data into knowledge. ”

VECTORS AND SCALARS

Mathematics
— for a Smarter
Tomorrow

1 Scalar

- A **scalar** is an individual numerical value, typically drawn from the real numbers ($\mathbb{R}$).
- It often represents the value of a **single attribute** such as age, temperature, salary, etc.
- Scalars are written using lowercase letters such as a , b , x .
- Examples:** 5, -3.2 , 0, 25, 100.

2 Vector

- A **vector** represents a given set of data as an ordered array of numerical values (scalars).
- The individual values in the array are called the **components** of the vector.
- The number of components is called the vector **dimension** (or length).
- Vectors are usually written using bold letters ($\mathbf{x}$) or as an array: $\mathbf{x} = [x_1, x_2, \dots, x_n]$.

Examples of Scalars (Attributes)

Age	Hourly Wage	Years of Experience	Temperature	Height (m)
25	30	5	36.5	1.75

Example: A 3-Dimensional Vector

A 3-dimensional vector representation of a 25-year-old person making 30 dollars an hour, and having 5 years of experience:

$$\mathbf{x} = [25, 30, 5]$$

$x_1 = 25$ (Age)
 $x_2 = 30$ (Hourly Wage)
 $x_3 = 5$ (Years of Experience)
Dimension: 3

More Examples

2-dimensional vector
(representing a point in 2D):

$$\mathbf{x} = [3, -1]$$

Dimension: 2

4-dimensional vector
(e.g., features of a house):

$$\mathbf{x} = [1200, 3, 2, 250000]$$

Dimension: 4

n-dimensional vector
(general form):

$$\mathbf{x} = [x_1, x_2, \dots, x_n]$$

Dimension: n

Scalars are individual numbers, while vectors are ordered collections of numbers.



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